# ATHAR HUSSAIN

Aspiring Data Scientist & Machine Learning Engineer

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## PROFESSIONAL SUMMARY

Aspiring Data Scientist & ML Engineer with a strong foundation in Machine Learning, Python, SQL, and Data Analytics, backed by 7 end-to-end projects spanning predictive modeling, customer analytics, database design, and business intelligence. Skilled in building and deploying models (XGBoost, Random Forest, Regression) with rigorous feature engineering, hyperparameter tuning, and evaluation. Currently expanding into Deep Learning, NLP, and MLOps to become a well-rounded AI Engineer capable of building production-grade AI systems. Maintains a 9+ SGPA while managing multiple concurrent projects.

## TECHNICAL SKILLS

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| **Languages** | Python (Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn), SQL (SQL Server, SSMS) |
| **Machine Learning** | Linear/Logistic Regression, Decision Trees, Random Forest, AdaBoost, Gradient Boosting, XGBoost, K-Means Clustering |
| **Model Development** | Feature Engineering, Hyperparameter Tuning (GridSearchCV, RandomizedSearchCV), Cross-Validation, Pipeline & ColumnTransformer, SHAP |
| **Evaluation Metrics** | $R^2$, MAE, RMSE, Accuracy, Precision, Recall, F1-Score, ROC-AUC |
| **Data Analysis & BI** | EDA, Data Cleaning, Statistical Analysis, Power BI, Business Intelligence Storytelling |
| **Deployment & Tools** | Streamlit, Joblib, Git & GitHub, Jupyter Notebook, VS Code, SQL Server Management Studio |

## PROJECTS

**Customer Churn Prediction** | *Python, Scikit-learn, XGBoost, SHAP, Streamlit*

- Built and compared 6 classification models (Logistic Regression, Decision Tree, Random Forest, AdaBoost, Gradient Boosting, XGBoost) to predict telecom customer churn.
- Tuned XGBoost via GridSearchCV to reach 79.7% accuracy and 0.848 ROC-AUC; applied SHAP for model explainability and deployed the model as an interactive Streamlit app.

**Customer Order SQL Project** | *SQL Server, SSMS, Python (Pandas, Faker)*

- Designed a relational database for a simulated Indian e-commerce business with 3 tables covering 200 customers, 40 products, and 1,484 orders, using Python to generate realistic synthetic data.
- Authored 85 SQL queries (JOINs, CTEs, Window Functions, Aggregations) to answer business questions on sales, customers, and product performance.

**Titanic: Comparative Classification Study** | *Python, Scikit-learn, Streamlit*

- Built an end-to-end ML pipeline with Pipeline/ColumnTransformer preprocessing, comparing Logistic Regression, Decision Tree, Bagging, and Random Forest for survival prediction.
- Tuned models with GridSearchCV and benchmarked on Accuracy, Precision, Recall, F1, ROC-AUC, and PR-AUC; deployed the top model via Streamlit.

**Retail Sales Data Analysis** | *Python, Pandas, Matplotlib, Seaborn*

- Cleaned and analyzed a retail dataset of 25,000 orders, 3,000 customers, and 250 products to surface business insights.
- Validated profit/discount business rules through EDA and identified top-performing categories, translating findings into a data-driven business narrative.

**House Price Prediction** | *Python, Scikit-learn (Ridge, Lasso, Elastic Net)*

- Engineered features with Pipeline/ColumnTransformer and built Linear, Ridge, Lasso, and Elastic Net regression models for housing price prediction.
- Achieved an $R^2$ score of 0.823 through hyperparameter tuning and cross-validation; prepared the model for Streamlit deployment.

**Titanic Survival Prediction** | *Python, Scikit-learn, Streamlit*

- Engineered features (FamilySize, IsAlone, Title extraction) to improve a Logistic Regression survival model, evaluated using ROC-AUC and other classification metrics.
- Deployed the trained model as an interactive Streamlit web application for real-time predictions.

**Customer Segmentation using K-Means** | *Python, Scikit-learn*

- Performed data cleaning, EDA, and IQR-based outlier detection, then scaled features with StandardScaler for clustering.
- Applied K-Means clustering with the Elbow Method and Silhouette Score to segment customers into actionable groups.

## EDUCATION

**B.Tech in Computer Science Engineering (CSE)** — **Expected Graduation: 2027**

*Shadan College of Engineering and Technology, Hyderabad* — **SGPA: 9+ (across 6 semesters)**

## SOFT SKILLS

Problem-Solving & Critical Thinking · Communication & Stakeholder Presentation · Collaboration & Teamwork · Self-Learning & Adaptability · Attention to Detail · Time Management